American Computer Science League

2024 Finals ● Short Problems Solutions ● Junior Division

1. Boolean Algebra

$AB + \overline{A} + B + \overline{A}\,\overline{B} = B(A + 1) + \overline{A}(1 + \overline{B})$
$= B + \overline{A}$

B. $\overline{A} + B$

2. Boolean Algebra

$\overline{A(\overline{B} + C)} + B(\overline{A + C}) + \overline{C}$
$= \overline{A} + \overline{\overline{B} + C} + B\,\overline{A}\,\overline{C} + \overline{C}$
$= \overline{A} + B\,\overline{C} + \overline{C}\,(\overline{A}B + 1)$
$= \overline{A} + B\overline{C} + \overline{C}$
$= \overline{A} + \overline{C}$

To be FALSE: $\overline{A} = 0$ and $\overline{C} = 0$. So $A = 1,\ C = 1$
Therefore, 2 make it FALSE. 1*1

A. 2

3. Bit-String Flicking

(RCIRC-3 01100) OR (NOT (10110 AND 10011))
= 10001 OR (NOT 10010)
= 10001 OR 01101
= 11101

D. 11101

4. Bit-String Flicking

Let X = abcde
LHS = (LCIRC-2 (10111 AND X))
= (LCIRC-2 (10111 AND abcde))
= (LCIRC-2 a0cde)
= cdea0

RHS = (RSHIFT-1 10100)
= 01010
LHS = RHS → cdea0 = 01010

So a = 1, b = *, c = 0, d = 1, e = 0. 1*010

B. 1*010

5. Recursive Functions

$f(18) = f(18-1) + 1 = f(17) + 1 = 10 + 1 = 11$
$f(17) = f(17+2) - 2 = f(19) - 2 = 12 - 2 = 10$
$f(19) = f(19+2) - 2 = f(21) - 2 = 14 - 2 = 12$
$f(21) = 21 - 7 = 14$
$f(11) = f(11+2) - 2 = f(13) - 2 = 6 - 2 = 4$
$f(13) = f(13+2) - 2 = f(15) - 2 = 8 - 2 = 6$
$f(15) = 15 - 7 = 8$
So $f(f(18)) = f(11) = 4$

A. 4

6. Recursive Functions

$f(13,7) = f(13-1,7+2) + 1 = f(12,9) + 1 = 0 + 1 = 1$
$f(12,9) = f(12-1,9+2) = f(11,11) + 1 = -1 + 1 = 0$
$f(11,11) = f(11-1,11+1) - 3 = f(10,12) - 3 = 2 - 3 = -1$
$f(10,12) = 12 - 10 = 2$

C. 1

7. Digital Electronics

The circuit translates to: $\overline{\overline{(A + B) + \overline{BC}} + C}$

$$\begin{aligned}\overline{\overline{(A + B) + \overline{BC}} + C} &= \overline{\overline{(A + B) + \overline{BC}} + C} \\ &= \overline{\overline{A + B}\,\overline{\overline{BC}} + C} \\ &= \overline{\overline{A}\,\overline{B}\,B\,C + C} \\ &= \overline{0 + C} \\ &= \overline{C}\end{aligned}$$

C. C

8. Digital Electronics

The Boolean expression represented by the following digital circuit is:

$$\overline{A(\overline{\overline{AB} + (B + C)}) + C}$$

$$\begin{aligned}\overline{A(\overline{\overline{AB} + (B + C)}) + C} &= \overline{A(\overline{\overline{AB}}\,\overline{B + C}) + C} \\ &= \overline{A\,A\,B\,\overline{B}\,\overline{C} + C} \\ &= \overline{0 + C} \\ &= \overline{C}\end{aligned}$$

This is TRUE when $C = 0$. Therefore **0 makes it TRUE.

C. 4

9. Prefix-Infix-Postfix

+ − * 2 − / * 6 2 3 1 ^ 2 3 5
= + − * 2 − / (* 6 2) 3 1(^ 2 3) 5
= + − * 2 − (/ (6 2 *) 3) 1(2 3^) 5
= + − * 2 (− (6 2 * 3/) 1)(2 3^) 5
= + − (* 2 (6 2 * 3 / 1 −))(2 3^) 5
= + (− (2 6 2 * 3 / 1 −*)(2 3^)) 5
= + (2 6 2 * 3 / 1 −*2 3^−) 5
= 2 6 2 * 3 / 1 −*2 3^−5+

B. 2 6 2 * 3 / 1 −*2 3^−5+

10. Prefix-Infix-Postfix

4 2 # 7 / 1 1 # 2 * − 5 9 # + = (4 2 #) 7 / (1 1 #) 2 * − (5 9 #) +
= (16−2) 7 / (1−1) 2 * − (25−9) +
= (14 7 /) (0 2 *) − 16 +
= (2 0 −) 16 +
= 2 16 + = 18

D. 18

11. Computer Number System

155414_{16} = 0001 0101 0101 0100 0001 0100_2
= 101 010 101 010 000 010 100_2
= 5 2 5 2 0 2 4_8
= 5252024_8

B. 5252024

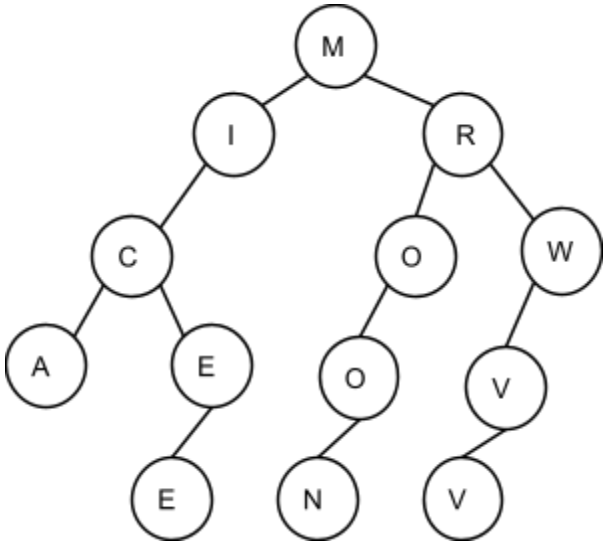
12. Computer Number Systems

$AAAA_{16} + BBB_{16} + CC_{16} + D_{16}$
= $1010101010101010_2 + 101110111011_2 + 11001100_2 + 1101_2$
= 1011 0111 0011 1110_2
= $B73E_{16}$

B. B73E

13. Data Structures

The binary search tree for MICROWAVEOVEN is:



The internal path length is: 2 * 1 + 3 * 2 + 4 * 3 + 3 * 4 = 2 + 6 + 12 + 12 = 32.

C. 32

14. Data Structures

Stack = 1 2 3 4 5, A=1, B=2, C=3, D=4, Stack = 2 5, E = 2, Stack = 1 3 5, F=1, G=3, Stack = 1 5, H=1, Stack = 4 5, J=4, Stack = 2 5, K=2. Therefore, Ball 2 is used by students B, E, and K. Ball 1 is used by A, F, and H. Ball 3 is used by C and G. Ball 4 is used by D and J. Ball 5 is never used.

B. B, E, K

15. Graph Theory

Amy → Tim → Deb → Sue → Joe

Tim → Deb → Sue → Joe → Amy or
Tim → Deb → Sue → Amy → Joe or
Tim → Sue → Joe → Amy → Deb

Sue → Joe → Amy → Tim → Deb

Joe → Amy → Tim → Sue → Deb or
Joe → Amy → Tim → Deb → Sue or
Joe → Amy → Deb → Sue → Tim or
Joe → Amy → Sue → Tim → Deb

Deb → Sue → Joe → Amy → Tim

D. All of them

16. Graph Theory

By inspection, there are none from vertex A and none from vertex E. The cycles are: BCB, BDCB, BDFCB, CDC, CDFC, CFC. There are 6 of them.

C. 6

17. What Does This Program Do? (Branching)

In order to put the 3 numbers in descending order, the first two if statements a<b and swap(a,b) work. After the first if statement, a < b and the second if statement b < c, b < a and c < b. Therefore, c is the largest. After the if statement a > b, the largest is in c, then b, and finally a.

B. a < b, swap(a,b)

18. What Does This Program Do? (Loops)

This gives the same number for val, sum1, and sum2 which is 28, a perfect number.
2^2 * (2^3−1) = 4 * 7 = 28
1 + 2 + … + (2^3−1) = 28
1^3 + 3^3 + … + (2^2−1)^3 = 28

B. 28

19. What Does This Program Do? (Arrays)

The first loop creates a 1D array of 4 values so it can be used as a frequency table to keep track of how many times each number in the given list occurs. The second loop tallies the results and the third loop finds the subscript (0-3) of the number that occurs most often. It then outputs the number that is found most often. The table is

how many 1s	how many 2s	how many 3s	how many 4s
num(0) = 1	num(1) = 2	num(2) = 4	num(3) = 3

Therefore, the output is 4.

D. 4

20. What Does This Program Do? (Strings)

This program creates two substrings of 5 bits and does an XOR evaluation of them. That produces the bit string 11100 for the given string "10101 01001".

A. 11100